The Visual Computer (2026) 42:52
https://doi.org/10.1007/s00371-025-04295-9

RESEARCH

PoseNorm-PCN: pose-normalized human point cloud completion from a single front view

Ming Li[1] · Guiqin Li[1] · Xihang Li[2]

Received: 14 August 2025 / Accepted: 23 November 2025 / Published online: 12 December 2025
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2025

Abstract
To address the challenges of pose variation sensitivity and limited detail recovery in human body point cloud completion, we propose PoseNorm-PCN, a pose normalization completion network designed for human body data. The method employs a three-stage modeling strategy. First, the pose normalization network transforms input point clouds with arbitrary poses into a canonical space. Second, the geometric offset inference network predicts z-axis displacements from the deformed front viewpoints to recover the back-side geometry. Finally, the recovered back point cloud in the canonical space is mapped back to the original pose through local coordinate transformations, yielding a complete human body point cloud consistent with the input pose. Extensive experiments on multiple real-world human scan datasets demonstrate that PoseNorm-PCN outperforms state-of-the-art completion methods, including PCN, PoinTr, and AdaPoinTr, achieving the lowest Chamfer Distance and Earth Mover's Distance. The model exhibits strong robustness and generalization under complex poses and diverse clothing conditions, making it highly suitable for three-dimensional human body reconstruction, virtual try-on, and human–computer interaction applications. The related code and resources have been released at: https://github.com/BestLiMing/PoseNorm_PCN.git.

Keywords Point cloud completion · Frontal human body scan · Pose normalization · Geometric offset · Inverse mapping

1 Introduction

Three-dimensional human body data reconstruction serves as a fundamental technology in numerous applications, including virtual try-on, behavioral analysis, and human–computer interaction. In practical scenarios, multi-view or full-body scanning systems can capture complete body geometry but are often constrained by high hardware costs, complex calibration, and synchronization requirements. In contrast, single-view scanning systems provide a more efficient and practicality alternative for capturing partial human geometry in real-world scenarios [1–3]. However, the self-occlusion of the back-side inevitably leads to missing geometric regions, significantly degrading the completeness and accuracy of the reconstructed 3D model. Therefore, completing the missing geometry from a single front view human point cloud becomes an indispensable step for achieving accurate and reliable 3D human reconstruction.

Despite the remarkable progress of point cloud completion methods based on deep learning [4–6], challenges persist in completing human-specific point clouds. The human body exhibits highly complex skeletal structures, large pose variations, and diverse inter-individual shapes, making it difficult for models to infer the occluded geometry of the back-side reliably. Articulated poses further cause geometric distortions and spatial inconsistencies among body parts, hindering the learning of a unified representation across samples [7]. Moreover, the inherent sparsity and non-uniform distribution of real-scanned point clouds, particularly in lateral body regions, make it difficult to recover fine-grained details. Existing approaches often treat completion as a global shape recovery task without explicit modeling of part-wise perceptual cues, which limits local geometric fidelity and leads to deviations in reconstructed body proportions.

✉ Guiqin Li
leeching@shu.edu.cn

1 Shanghai Key Laboratory of Intelligent Manufacturing and Robotics, Shanghai University, No. 333, Nan Chen Road, Bao Shan District, Shanghai 200444, China

2 College of Mechanical and Automotive Engineering, Ningbo University of Technology, No. 201, Fenghua Road, Jiangbei District, Ningbo 315211, Zhejiang, China

To address these challenges, we propose PoseNorm-PCN, a pose-normalized human body point cloud completion network designed for single-view frontal scans. First, the input point cloud is transformed from its original pose space to a unified T-pose canonical space, which effectively eliminates geometric distortions and spatial inconsistencies caused by pose variation and self-occlusion. Then, in this canonical space, the completion problem is reformulated as a one-dimensional depth-offset regression along the z-axis, where the network predicts per-point offsets from the visible front geometry to the missing back surface. This conversion simplifies the task from full 3D inference to compact 1D modeling while preserving geometric continuity. Finally, the completed back-side points are mapped back to the original pose space through a local coordinate inverse transformation, ensuring that the reconstructed human body point cloud remains structurally consistent with the observed input. The main contributions are as follows:

(1) A pose normalization strategy is proposed to map the input point cloud into a canonical T-pose space, alleviating geometric distortion and spatial inconsistency caused by complex postures;
(2) The completion task is reformulated as a one-dimensional depth-offset regression along the canonical z-axis, simplifying the problem from full 3D inference to compact 1D modeling while preserving geometric continuity;
(3) A local inverse mapping mechanism is designed to restore the completed geometry to the original pose space, ensuring spatial consistency and structural integrity.

2 Related words

2.1 Human body point cloud completion

In front view scanning scenarios, the human body surface often contains numerous self-occluded regions caused by clothing and limited viewpoints, resulting in substantial loss of geometric and topological information. This incomplete observation introduces ambiguity in 3D human reconstruction, which becomes particularly challenging when multi-view data are unavailable, making it difficult to recover realistic human shapes from limited inputs.

Point clouds, as a lightweight and unstructured 3D representation, provide lower computational complexity while retaining fine-grained geometric details, making them suitable for real-time 3D perception systems. However, front view point clouds from a single viewpoint are typically severely incomplete, posing significant challenges for data-driven methods to accurately infer complete human shapes, often leading to topological ambiguities and distortions in local details. Consequently, point cloud completion has become a crucial step in enhancing 3D human reconstruction quality.

Deep learning-based point cloud completion methods have achieved notable progress. PCN [8] pioneered an end-to-end framework that fuses local geometric features with global shape priors to generate complete point clouds. MISO-PCN [9] introduced a mutual information–maximization similarity operation, improving structural consistency and realism, but no publicly available code has been provided for reproduction. FinerPCN [10] adopts a coarse-to-fine two-stage strategy with point-wise convolutions, enhancing local details recovery while maintaining global coherence. However, its code is also unavailable for reproducibility. With the advent of Transformer architectures, several methods have leveraged their capabilities for point cloud completion [11]. PoinTr [12] represents points as structured "point token" and applies geometry-aware self-attention to generate complex topologies. AdaPoinTr [13] further incorporates adaptive geometry-aware mechanisms and dynamic feature learning, improving completion quality, efficiency, and model generalization.

2.2 Dense surface correspondences

Establishing dense surface correspondences between the original spatial point cloud and a normalized spatial point cloud is essential for geometric consistency and semantic alignment in human body point cloud completion. Such correspondences help capture structural distributions under complex poses and provide reliable shape priors and topological constraints for inferring missing regions.

Early work, such as DensePose [14], employed a two-branch network for classification and regression to map image pixels to local coordinates on a 3D template mesh. While pioneered dense correspondences from 2D images to standard human body models, it relies on predefined geometric priors and discretized outputs, which may introduce seam artifacts between body parts. HumanGPS [15] introduced topological stability under large pose variations or occlusions by aligning Euclidean distances in feature space with geodesic distances on the 3D surface. Other studies focus on shape and pose invariant correspondences [16–19], aiming to achieve consistent matching across individuals and poses. These approaches are often used in parametric model fitting or non-rigid registration and typically require initialization close to the ground truth to ensure optimization stability, making them less robust for highly deformed or severely incomplete scans.

Recently, Lunscher et al. [20] proposed a method for SMPL registration in which dense surface correspondences are established between the SMPL model and scanned point clouds. This approach allows precise alignment across the entire body and ensures geometric consistency of the registered model. Building on this idea, Li et al. [21] introduced the VSMPL model, which maps scanned point clouds in arbitrary poses to a canonical T-pose, enabling inference of the underlying human shape beneath clothing. AniSym-Net [22] further exploits bilateral symmetry and pose-dependent features to enhance back surface reconstruction and reduce artifacts under occlusion. By leveraging these dense correspondences between original and normalized point clouds, these methods provide robust geometric constraints and reliable shape priors, which are crucial for human body shape estimation and completion of missing regions under complex poses.

In the context of point cloud completion, establishing accurate dense correspondences between original and canonical space point clouds is particularly beneficial. It enables more stable geometric reasoning, better preservation of local structures, and improved robustness when handling extreme poses or missing data.

3 Method

We present a method for complete human body geometry completion from a single front view point cloud, comprising three main stages: (1) pose normalization of the input point cloud; (2) geometric offset inference based on the front view to recover the back-side structure; (3) mapping the completed points back to the original pose space via local coordinate systems. The overall framework is shown in Fig. 1.

In the first stage, the pose normalization network transforms the input front view point cloud into a canonical space, effectively reducing pose variability and enabling more consistent reasoning of the occluded back-side geometry. In the second stage, the geometric offset inference network predicts geometric offsets for the missing regions within this canonical space, generating a complete human point cloud. In the final stage, the completed point cloud is inverse mapped to the original space by establishing point-wise correspondences and leveraging local coordinate systems between the canonical and original spaces.

3.1 Pose normalization modeling

To reduce the complexity introduced by varying human poses in point cloud completion, a pose normalization network is employed to transform human body point clouds from arbitrary poses into a unified canonical T-pose space. This process eliminates non-rigid influences such as soft-tissue deformation, self-occlusion, and limb overlapping, thus simplifying the subsequent geometric reasoning task. The input of this stage is a front view human point cloud $\mathcal{P}_F \in \mathbb{R}^{N\times 3}$, and the output is its normalized counterpart $\mathcal{P}_{F,t\text{pose}} \in \mathbb{R}^{N\times 3}$ in canonical space.

In complex poses with interlaced or highly bent limbs, global deformation fields often fail to establish accurate correspondences between the original and canonical spaces. To enhance mapping stability, slight adjustments are introduced to the canonical T-pose, as shown in Fig. 2. Specifically, the leg pose is adjusted to form a 30° angle between the legs, increasing limb distinguishability without inducing excessive deformation. The hands are rotated 90° outward so that the palms face forward, improving visibility in the hand region and mitigating local occlusions. These refinements

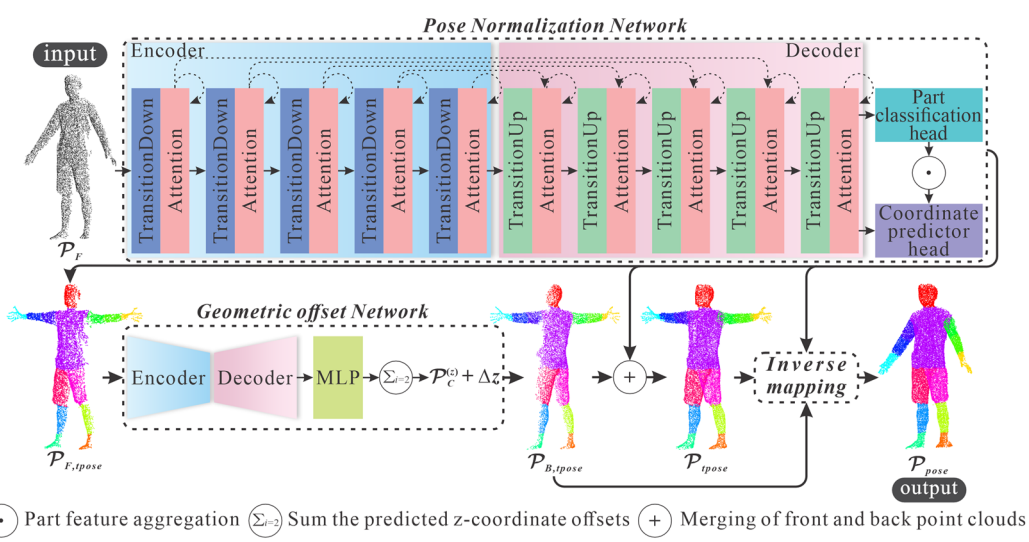

Fig. 1 The proposed method framework

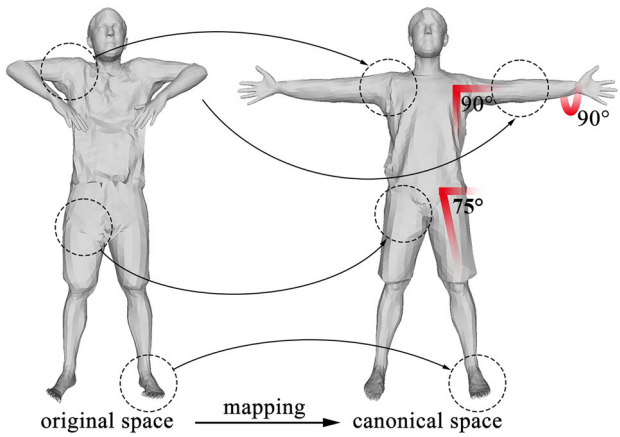

Fig. 2 Canonical T-poses definition

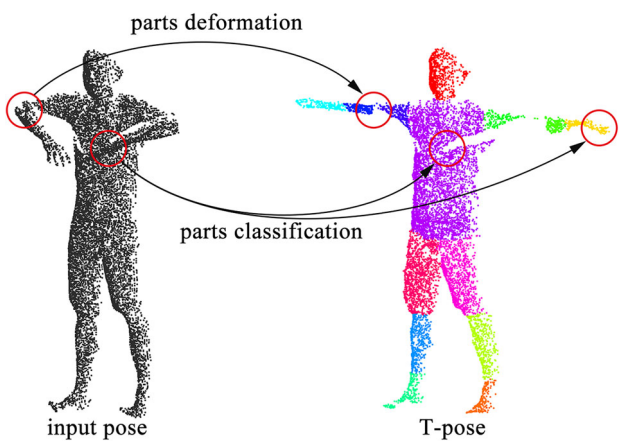

Fig. 3 Dense correspondences modeling

improve the geometric separability of body parts and provide clearer structural cues for subsequent completion.

The pose normalization network adopts a part-aware multi-branch collaborative optimization strategy, as shown in Fig. 3. First, a PointTransformer encoder extracts multi-scale geometric features from the input point cloud. Then, two parallel branches are introduced: (1) a part classification branch, which predicts the semantic probability distribution over $K = 14$ predefined body parts using a Softmax layer; and (2) a coordinate predictor branch, which estimates target coordinates in the canonical space via a multi-layer perceptron (MLP). Finally, a part feature aggregation mechanism integrates the deformation results according to semantic probabilities, generating the final normalized point cloud. This collaborative strategy dynamically balances the influence of different parts and significantly improves the accuracy and robustness of spatial mapping between the original and canonical spaces.

Given an input pose point cloud $\mathcal{P}_F = \{p_i \in \mathbb{R}^3\}_{i=1}^N$, with $N = 10{,}000$, the corresponding points in the canonical space, denoted as $\mathcal{P}_{F,t\text{pose}} = \{c_i \in \mathbb{R}^3\}_{i=1}^N$ are obtained by solving:

$$c_i = \sum_{k=1}^{14} w_{ik}(p_i;\theta) \cdot \mathcal{T}_k(p_i;\theta) + \varepsilon_i \tag{1}$$

where c_i is the target coordinate of point p_i in the normalized space. $\mathcal{T}_k(\cdot;\theta)$ represents the sub-deformation field for the k-th part, and $w_{ik}(p_i;\theta)$ is the probability weight indicating the membership of point p_i to part k, computed as $w_{ik} = \text{Softmax}\ (\text{f}(x_i;\theta))_k$. Here, x_i is the feature vector of point p_i, θ, denotes the network parameters, and ε_i is the residual term.

3.2 Geometric offset method based on front view point cloud

Following the pose normalization of front view point clouds, the back-side geometry completion task can be greatly simplified. In the canonical space, where limb arrangements are orderly and topological consistency is preserved, the task reduces to predicting per-point offsets along the z-axis from the visible front surface to the occluded back surface. However, challenges remain due to self-occlusions, clothing-induced deformations, and individual shape variations, which introduce ambiguities in the observed geometry.

To address these challenges, we propose a geometric offset inference network that reconstructs back-side geometry from a pose-normalized front view point cloud. The input of this stage is the front view point cloud $\mathcal{P}_{F,t\text{pose}}$, and the output is a set of z-axis offsets $\{\Delta z_i\}_{i=1}^N$. The coordinates of the completed back points are computed as $p_i{\prime} = (x_i, y_i, z_i + \Delta z_i)$.

The network adopts a single-branch encoder–decoder architecture. A PointTransformer-based encoder extracts multi-scale geometric features, capturing both local detail and global semantic context. These features are fed into a lightweight MLP decoder, which regresses per-point z-axis offsets. By working in the canonical space, the network avoids complex non-rigid registration and reduces ambiguities caused by pose variations, allowing efficient and accurate back-side reconstruction (Fig. 4).

Unlike symmetry-based methods, our approach does not rely on strict geometric symmetry assumptions. Instead, it learns a nonlinear mapping from front view observations to back geometry in an end-to-end manner, accommodating variations in individual shape, pose, and clothing. This design improves robustness and generalization, making it suitable for real-world scenarios.

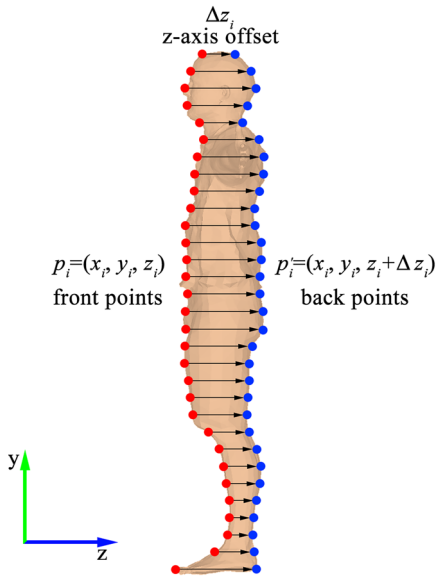

Fig. 4 Geometric offset inference

3.3 Inverse mapping of the completed point cloud using local coordinate systems

After completing the back point cloud in the canonical space, it is necessary to restore it to the original space to recover pose-specific geometric details. To achieve this, we propose an inverse mapping strategy based on local coordinate systems.

Let $\mathcal{P}_{F,t\text{pose}} = \{c_i \in \mathbb{R}^3\}_{i=1}^N$ denote the front point cloud in the canonical space, and $\mathcal{P}_{B,t\text{pose}} = \{q_i \in \mathbb{R}^3\}_{i=1}^N$ denote the corresponding completed back points. For each point pair, define an offset vector $\Delta d_\text{i} = \text{q}_\text{i} - \text{c}_\text{i}$, representing the geometric displacement from the front surface to the back in the canonical space.

To transfer these offsets back to the original space while preserving local structure, we compute a local rigid transformation (R, t) for each point. Specifically, for each front point, we calculate its normal vectors n_i and $\text{n}_{\text{i,tpose}}$ in the original and canonical spaces. These normals define local coordinate frames that guide the structure-preserving transformation. The rotation R is obtained via Procrustes alignment. Let p_i and c_i denote corresponding front points in the original and canonical spaces. The covariance matrix H is defined as:

$$H = \frac{1}{N}\sum_{i=1}^{N}\left(c_i - \mu_{t\text{pose}}\right)\left(p_i - \mu_\text{pose}\right)^T \tag{2}$$

where $\mu_\text{pose}=\frac{1}{\text{N}}\sum_{\text{i}=1}^{\text{N}}\text{p}_\text{i}$ and $\mu_\text{tpose}=\frac{1}{\text{N}}\sum_{\text{i}=1}^{\text{N}}\text{c}_\text{i}$ are the centroids of the two point sets.

Performing singular value decomposition (SVD) on H, i.e., $[\text{U, S, V}^\text{T}] = \text{SVD(H)}$, yields the rotation $\text{R}=\text{VU}^\text{T}$. To prevent reflection, if $\det(\mathbf{R}) < 0$, the last column of V is negated and R is recomputed. The translation vector is computed as:

$$\mathbf{t} = \mu_\text{pose} - \mathbf{R}\mu_{t\text{pose}} \tag{3}$$

Finally, the back points are reconstructed in the original space as:

$$q_i = p_i + \mathbf{R}\cdot\Delta\mathbf{d}_i \tag{4}$$

This local transformation strategy ensures that the back surface geometry, completed in the canonical space, is accurately restored to the original space. Compared to global affine transformations, it better preserves fine-grained structural details, enhancing the realism and anatomical full-body point cloud.

4 Dataset construction

We construct training samples for human pose normalization and point cloud completion based on the CAPE dataset [23], which contains 15 subjects (10 males and 5 females) across more than 600 motion sequences. Each sequence provides a clothed human mesh along with a corresponding undressed SMPL body mesh. The SMPL pose and shape parameters are also available, enabling accurate modeling of pose transformations and body shape variations.

Our goal is to create a point cloud dataset that is consistent in scale, spatially aligned, and exhibits front and back correspondences. This dataset supports two primary learning tasks: (1) Pose normalization, which maps posed human meshes into a canonical space; (2) point cloud completion, which predicts the back geometry of the human body from front view observations. This resulting unified dataset enables joint training and evaluation of models for both pose alignment and geometric reconstruction.

4.1 Pose normalization ground truth generation

Given an input mesh $\mathcal{M} = (V,\ F)$ with vertices V and faces F, along with its associated SMPL pose parameters θ, we apply linear blend skinning (LBS) to deform the mesh into the canonical T-pose:

$$\mathcal{M}_{t\text{pose}} = \mathcal{T}(\mathcal{M}, \theta) \tag{5}$$

where $\mathcal{T}(\cdot)$ denotes the LBS-based pose normalization function. The resulting mesh $\mathcal{M}_{t\text{pose}}$ serves as a canonical reference in the canonical space.

To simulate realistic partial observations, we render a depth map of the posed mesh $\mathcal{M}$ from a frontal camera

using PyTorch3D. From this depth map, we randomly sample $H = 10,000$ points to construct the front view point cloud. Then, each sampled point $p_i \in \mathcal{P}_F$ is mapped to the canonical space as follows: (1) Project p_i onto its nearest triangle on the posed mesh $\mathcal{M}$ to obtain the closest surface point $\widetilde{p}_i$; (2) compute the barycentric coordinates $(\lambda_{i1}, \lambda_{i2}, \lambda_{i3})$ of $\widetilde{p}_i$ with respect to the triangle's vertices $v_{i1}, \mathbf{v}_{i2}, \mathbf{v}_{i3}$; (3) use the corresponding triangle in the normalized mesh $\mathcal{M}_{t\text{pose}}$ (i.e., with the same face index) to map to the canonical space:

$$q_i = \lambda_{i1}\mathbf{V}_{i1} + \lambda_{i2}\mathbf{v}_{i2} + \lambda_{i3}\mathbf{v}_{i3} \tag{6}$$

This procedure yields the normalized front view point cloud $\mathcal{P}_{F,t\text{pose}} = q_i{}_{i=1}^{K}$, which serves as the supervision target for the pose normalization network. Each input-target pair $(\mathcal{P}_F, \mathcal{P}_{F,t\text{pose}})$ thus forms a supervised training sample for network learning.

4.2 Point cloud completion supervision

Given a normalized front view point cloud $\mathcal{P}_{F,t\text{pose}}$, we cast a ray from each point q_i along the positive z-axis:

$$\ell(t) = q_i + t \cdot \mathbf{e}_z \tag{7}$$

where $\mathbf{e}_z = [0,0,1]^\top$ is the unit vector along the z-direction.

We compute the intersection points between this ray and the normalized mesh $\mathcal{M}_{t\text{pose}}$. Owing to the symmetric and upright nature of the T-pose, the last intersection point along the ray, denoted as b_i, is defined as the ground truth back point corresponding to q_i. The depth displacement along the z-axis between the front and back points is computed as:

$$\Delta z_i = z(b_i) - z(q_i) \tag{8}$$

where $z(\cdot)$ denotes extraction of the z-coordinate. The resulting scalar Δz_i serves as the supervision signal for learning geometric completion in the normalized space.

During inference, the network predicts a depth-offset $\Delta\hat{z}_i$ for each point q_i, from which the estimated back point is reconstructed as:

$$\hat{b}_i = q_i + \Delta\hat{z}_i \cdot \mathbf{e}_z \tag{9}$$

By combining the observed front points q_i with the reconstructed back points $\widehat{b}_i$, we obtain a completed and geometrically consistent point cloud in the canonical space.

5 Training details

The pose normalization network and the geometric offset inference network are trained separately in a supervised manner, with each network optimized for its specific task. Both networks are trained using the Adam optimizer with an initial learning rate of 1×10^{-3} and a batch size of 10. Training is performed for 500 epochs to ensure sufficient convergence. In practice, the pose normalization network requires approximately 83 h to train, while the geometric offset inference network takes about 67 h.

To enhance the generalization capability of the pose normalization network, several data augmentation strategies are employed during training, including random rotations, translations, normal vector perturbations, and point-level jittering. In contrast, no data augmentation is applied to the geometric offset inference network, as its input relies on precise geometric offset annotations. Omitting augmentation in this case prevents the introduction of errors caused by inaccurate transformations.

5.1 Loss function definition for posture normalization network

The pose normalization network aims to transform point clouds from arbitrary poses into a canonical space while preserving the semantic consistency of body parts. To achieve this, a multi-task loss function is employed, consisting of two components: the body part classification loss (L_{part}) and the geometric correspondence loss (L_{pose}), which jointly enforce semantic consistency and geometric alignment.

To maintain semantic consistency between the normalized point cloud and the original input, a cross-entropy loss is applied to supervise the predicted body part labels against the ground truth annotations:

$$\mathcal{L}_{\text{part}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik}\log(p_{ik}) \tag{10}$$

where N denotes the number of sampled points, $K = 14$ is the number of body part categories, y_{ik} is the ground truth label indicating whether the i-th category, and p_{ik} is the model-predicted probability of the point belonging to that class.

To improve geometric alignment accuracy in 3D space, we employ the mean squared error (MSE) to measure the coordinate differences between the predicted and target point clouds in the canonical space:

$$\mathcal{L}_{\text{pose}} = \frac{1}{N}\sum_{i=1}^{N}(a_{\text{pred}}[i] - a_{\text{target}}[i])^2 \tag{11}$$

where $a_{pred}[i]$ and $a_{target}[i]$ denote the predicted and ground truth coordinates of the i-th point in the standardized space.

Finally, the overall loss function for the pose normalization network is defined as the weighted sum of these two terms:

$$\mathcal{L}_{PN} = \mathcal{L}_{part} + \lambda_1 \cdot \mathcal{L}_{pose} \tag{12}$$

where the weighting factor λ_1 balances the contributions of the classification loss and the geometric alignment loss during training. In our experiments, we set $\lambda_1 = 1000$, emphasizing the importance of geometric alignment while maintaining strong semantic consistency.

By jointly optimizing these objectives, the pose normalization network achieves accurate and semantically consistent pose normalization, providing high-quality inputs for the subsequent point cloud completion task.

5.2 Loss function definition for geometric offset network

The geometric offset inference network adopts a single-branch architecture, aiming to directly predict the z-axis displacement for each back-side point based on the front view point cloud in the canonical space. The input is the set of front points $\mathcal{P}_{F,t\text{pose}} = \{q_i \in \mathbb{R}^3\}_{i=1}^{N}$ and the network outputs the corresponding z-axis offset Δz_i for each point, which are then used to reconstruct the back point coordinates.

To optimize the network, we employ the mean squared error (MSE) loss to measure the discrepancy between the predicted offset and the ground truth offsets:

$$\mathcal{L}_{offset} = \frac{1}{N}\sum_{i=1}^{N}(\hat{z}_i - \Delta z_i)^2 \tag{13}$$

where $\hat{z}_i$ denotes the model-predicted z-axis offset for the i-th point and Δz_i is the corresponding ground truth offset obtained through ray intersection detection.

This loss effectively guides the network to learn accurate depth-wise displacements from front points to their back counterparts, ensuring high-fidelity reconstruction of the complete 3D human shape.

6 Experiment

To comprehensively evaluate the performance of the proposed method in both the pose normalization network and the geometric offset inference network, we design a series of systematic experiments from multiple perspectives. First, an ablation study based on part feature aggregation is conducted to verify its impact on pose normalization process, allowing us to assess how semantic guidance improves alignment accuracy and structural consistency. Second, another ablation experiment is performed by removing the pose normalization module entirely to investigate how direct completion on non-normalized pose point clouds affects geometric reconstruction performance. This comparison helps reveal the necessity of pose normalization in simplifying the completion task and maintaining geometric stability. Third, the overall effectiveness and generalization capability of the framework are evaluated on a test set containing 1200 samples that cover a wide range of body shapes and poses, demonstrating the robustness and stability of the method under varying conditions. Fourth, a comparative experiment is conducted to evaluate the proposed pose normalization network against VSMPL [21], quantifying its alignment accuracy under complex pose variations. Finally, to assess the performance of the point cloud completion strategy, we conduct comparative experiments on real-scanned human data from the THuman2.0 [24], BUFF [25], and Multi-Garment [26] datasets. Our method is compared with several state-of-the-art point cloud completion approaches, including PCN [8], PointTr [12], and AdaPoinTr [13], providing a comprehensive evaluation across different reconstruction paradigms in the context of human body completion tasks.

The evaluation metrics employed are the Chamfer Distance (CD) and Earth Mover's Distance (EMD). The CD measures the average minimum squared euclidean distance between points in the predicted point cloud S_1 and the target point cloud S_2, reflecting the overall geometric similarity:

$$d_{CD}(S_1, S_2) = \frac{1}{S_1}\sum_{x\in S_1}\min_{y\in S_2}\|x-y\|_2^2 + \frac{1}{S_2}\sum_{y\in S_2}\min_{x\in S_1}\|y-x\|_2^2 \tag{14}$$

The EMD evaluates the minimal transport cost to optimally match all points from one set to another, providing a more precise measure of global structural alignment:

$$d_{EMD}(S_1, S_2) = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{ij}d_{ij}}{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{ij}} \tag{15}$$

where f_{ij} denotes the flow from i-th point to j-th point, and d_{ij} is the Euclidean distance between them.

In addition to these shape similarity metrics, we report inference time to quantify the computational efficiency of the model.

Table 1 The impact of part feature aggregation module on pose normalization modeling

Methods	CD	EMD
Weighting	**10.85**	**4.49**
No weighting	10.98	6.70

The best result for each metric is presented in bold

Table 2 Comparison of point cloud completion with and without pose normalization module

Methods	CD	EMD
With the pose normalization module	**10.48**	**4.20**
Without the pose normalization module	11.73	6.11

The best result for each metric is presented in bold

6.1 Impact analysis of part feature aggregation

To evaluate the contribution of the proposed part feature aggregation module in pose normalization modeling, we train two variants of the pose normalization network: one incorporating body part probability weighting during feature fusion (weighting) and another without this weighting mechanism (no weighting). All other architectures settings and training configurations remain identical to ensure fair comparison.

As shown in Table 1, introducing the part feature aggregation yields consistent improvements across both CD and EMD metrics. Specifically, CD decreases from 10.98 to 10.85, indicating enhanced global shape reconstruction, while the EMD shows a more pronounced reduction of 2.21, demonstrating improved point-wise correspondence and spatial alignment accuracy. The qualitative results in Fig. 5 further support these findings, where visual inspection reveals that the part feature aggregation better preserves local deformations, particularly in challenging regions such as the arms and legs, leading to more accurate and stable pose alignment.

In summary, embedding semantic body part information into the feature aggregation process enables the network to more effectively model the intricate relationships between local geometry and global pose. This enhancement not only improves pose normalization accuracy but also strengthens the reliability and fidelity of subsequent point cloud completion, ensuring more coherent and anatomically consistent reconstructions.

6.2 Point cloud completion performance analysis without the pose normalization module

The effectiveness of the pose normalization stage is further evaluated through an ablation experiment in which the pose normalization network is removed. In the setting, the posed front view point cloud is directly fed into the geometric offset inference network for completion. All other training configurations and parameters remain unchanged to ensure a fair comparison.

Table 2 summarizes the quantitative results of point cloud completion with and without the pose normalization module. The results show a notable decline in reconstruction accuracy after the module is removed: the CD increases from 10.48 to 11.73, and the EMD rises from 4.20 to 6.11. This consistent degradation across both metrics indicates that the absence

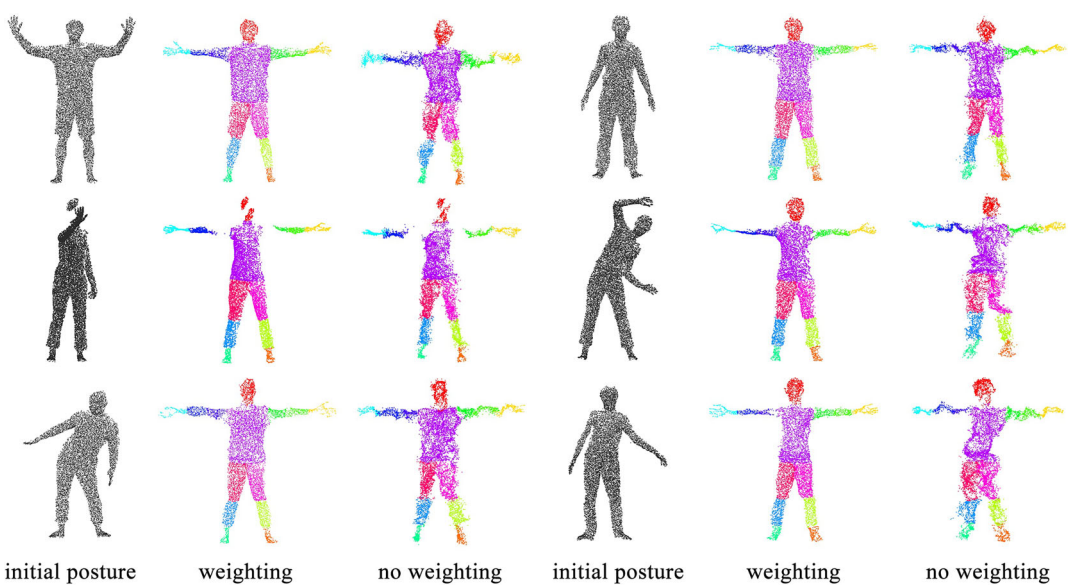

Fig. 5 Results of pose normalization modeling

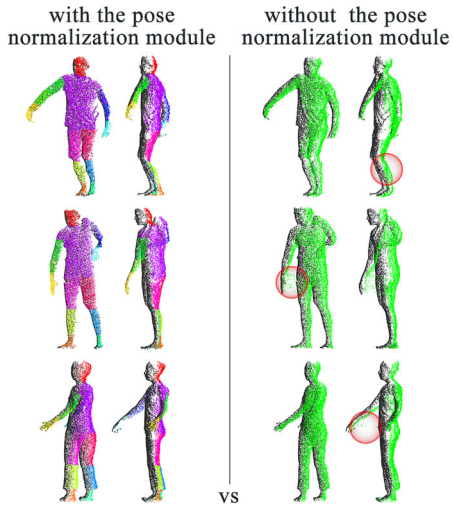

Fig. 6 Comparison of completion results with and without pose normalization

of pose normalization leads to misaligned geometric correspondences and degraded structural consistency between front and back regions.

Figure 6 shows the qualitative comparison of point cloud completion results with and without the pose normalization module. It can be clearly seen that the model with pose normalization produces more accurate and continuous geometry under complex poses, while the version without normalization exhibits structural distortion and missing surface regions.

The performance degradation mainly arises from the lack of a unified spatial reference, which leads to several geometric inconsistencies. Without pose normalization, limb articulation causes rotation of local coordinate axes, resulting in inconsistent depth references and unstable spatial reasoning. Meanwhile, poses such as arm raising or forward leaning introduce depth overlaps between the front and back surfaces, leading to geometric interpenetration and discontinuities. Furthermore, directional inconsistencies among different body segments hinder the propagation of semantic consistency constraints, causing local drift and misalignment across body regions.

6.3 Geometric offset inference network performance evaluation

To evaluate the effectiveness of the geometric offset inference network in reconstructing back-side point clouds, we conduct quantitative evaluations on the test set. The front view point cloud in the canonical space is fed into the network to directly predict the back-side geometry.

As shown in Table 3, the proposed network achieves favorable results across all metrics. Low CD and EMD scores indicate high fidelity of the reconstructed geometry relative to the ground truth, while the short inference time (0.12 s) demonstrates potential for real-time applications. Representative visual results are presented in Fig. 7.

Table 3 Performance evaluation of geometric offset network based on test set samples

Method	CD	EMD	Time
Geometric offset network	4.03	5.12	0.12 s

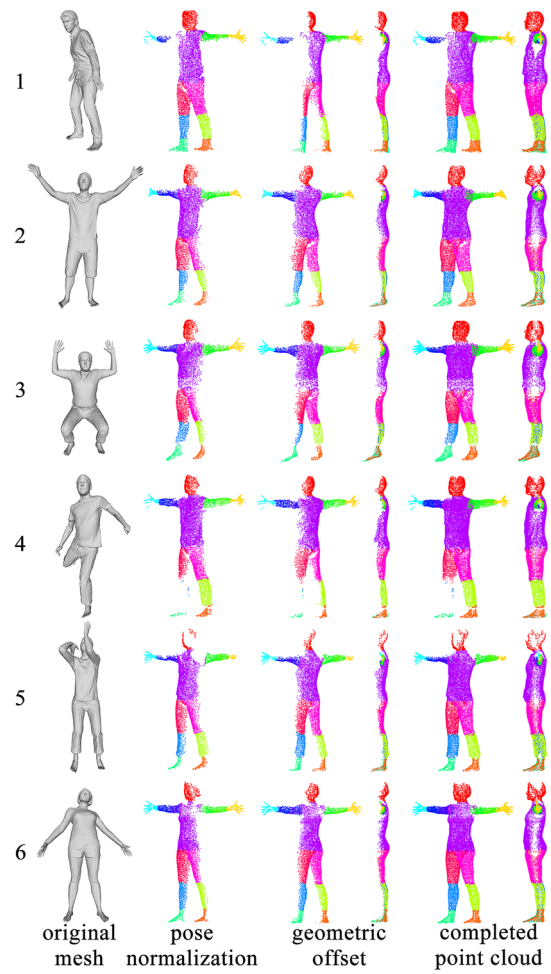

Fig. 7 Geometric offset inference results

As shown in Fig. 7, the network effectively reconstructs the back-side geometry despite partial missing regions in the input point cloud caused by self-occlusion in the original pose. The torso and limb regions exhibit strong structural integrity and spatial coherence, and the reconstructed contours closely match the ground truth, confirming the network's ability to recover global pose and overall shape.

However, certain fine-scale details, such as spinal curvature, gluteal folds, and scapular protrusions, are smoothed out in the reconstructed surfaces. Local deformations caused

by clothing wrinkles or fabric draping are also incompletely recovered, resulting in slightly flattened appearances under complex garment conditions. This limitation mainly stems from the network's unidirectional z-axis prediction design, which focuses on depth-wise displacements and does not explicitly capture full 3D surface variations. Additionally, since the network relies solely on front view observations, back-side geometry is inferred from learned z-axis offsets. While this generally produces plausible reconstructions, it leads to sparsity along lateral body regions and overly smooth surface. Although the smoothness enhances reconstruction stability, it limits the recovery of fine-scale geometric details in areas that are not directly observed.

6.4 Pose normalization performance evaluation

The accuracy of pose normalization is critical for the subsequent point cloud completion. To verify and analyze the performance of the proposed pose normalization network, we conduct a comparative experiment against the VSMPL method [21]. The experiment aims to quantify alignment accuracy under complex poses and evaluate its impact on subsequent point cloud completion.

As shown in Fig. 8, front view point clouds are first input into both methods to generate normalized pose representations. Despite variations in pose and severe self-occlusion, our method produces more complete and semantically consistent results. Compared with VSMPL, which is originally designed for internal body pose normalization based on parametric models, our approach demonstrates superior robustness when applied to clothed human scans. The results show that VSMPL tends to exhibit noticeable distortions and discontinuities, especially around the limbs and torso, where clothing-induced geometric deviations break the underlying body-skin correspondence assumptions.

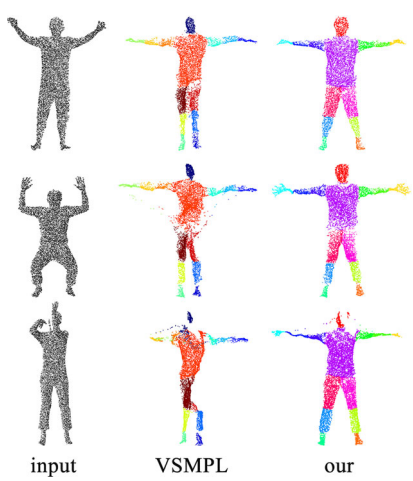

Fig. 8 Comparison of pose normalization results between the proposed method and VSMPL

In contrast, our pose normalization network effectively aligns the torso and limbs into a canonical configuration while maintaining structural coherence and local continuity. The normalized outputs exhibit smoother transitions across joints and more accurate part-level alignment. This improvement facilitates downstream point cloud completion by providing stable pose-consistent input geometry, reducing ambiguity and geometric artifacts introduced by complex clothing and occlusion.

6.5 Point cloud completion performance evaluation

To evaluate the effectiveness, robustness, and generalization of the proposed method in human point cloud completion, we conduct comprehensive comparisons against several state-of-the-art approaches on the test set. The selected baselines represent distinct mainstream paradigms: PCN [8], a classic encoder–decoder framework; PointTr [12], a Transformer-based generation model; and AdaPoinTr [13], an adaptive feature extraction framework that dynamically infers missing regions from contextual cues. For fairness, all baseline models are evaluated using the official implementations and pretrained parameters released by the original authors. Together, these baselines cover representative designs based on CNNs, attention mechanisms, and adaptive reasoning.

It is worth noting that the input point clouds contain missing regions mainly caused by self-occlusion in the frontal scans, which makes the completion task particularly challenging.

As shown in Table 4, our method achieves the lowest CD, indicating superior capability in producing geometrically accurate and structurally complete human point clouds. Although the EMD is slightly higher than that of AdaPoinTr, this can be attributed to our front-to-back inference strategy: when input exhibit significant occlusion or sparsity, the quality of the reconstructed back surface may degrade, leading to a higher optimal transport cost between point sets.

To further assess qualitative performance, Fig. 9 presents visual comparisons across methods. Our approach excels at preserving the structural integrity of human shapes, particularly in anatomically complex regions such as limb-torso connections. The reconstructed geometry is natural and continuous, avoiding artifacts like surface penetration or misalignment. While certain finer details (e.g., spine, shoulders) may appear slightly smoother than in the ground truth, the overall geometry aligns closely with the target geometry.

The reconstruction quality also reflects the characteristics of the input. For samples with clearly separated limbs and minimal occlusion (e.g., samples 4 and 6), our network produces topologically consistent and high-fidelity shapes by leveraging both local geometric cues and global human body priors. In more challenging samples with severe self-occlusion (e.g., sample 5), overlapping structures and

Table 4 Performance comparison of point cloud completion methods on the test set

Methods	CD	EMD	Parameters	Time
PCN	20.83	31.76	**6.52 M**	6.52 s
PoinTr	11.03	25.36	42.50 M	7.05 s
AdaPoinTr	9.36	**14.76**	32.49 M	6.71 s
Our	**7.85**	20.26	7.84 M	**0.14 s**

The best result for each metric is presented in bold

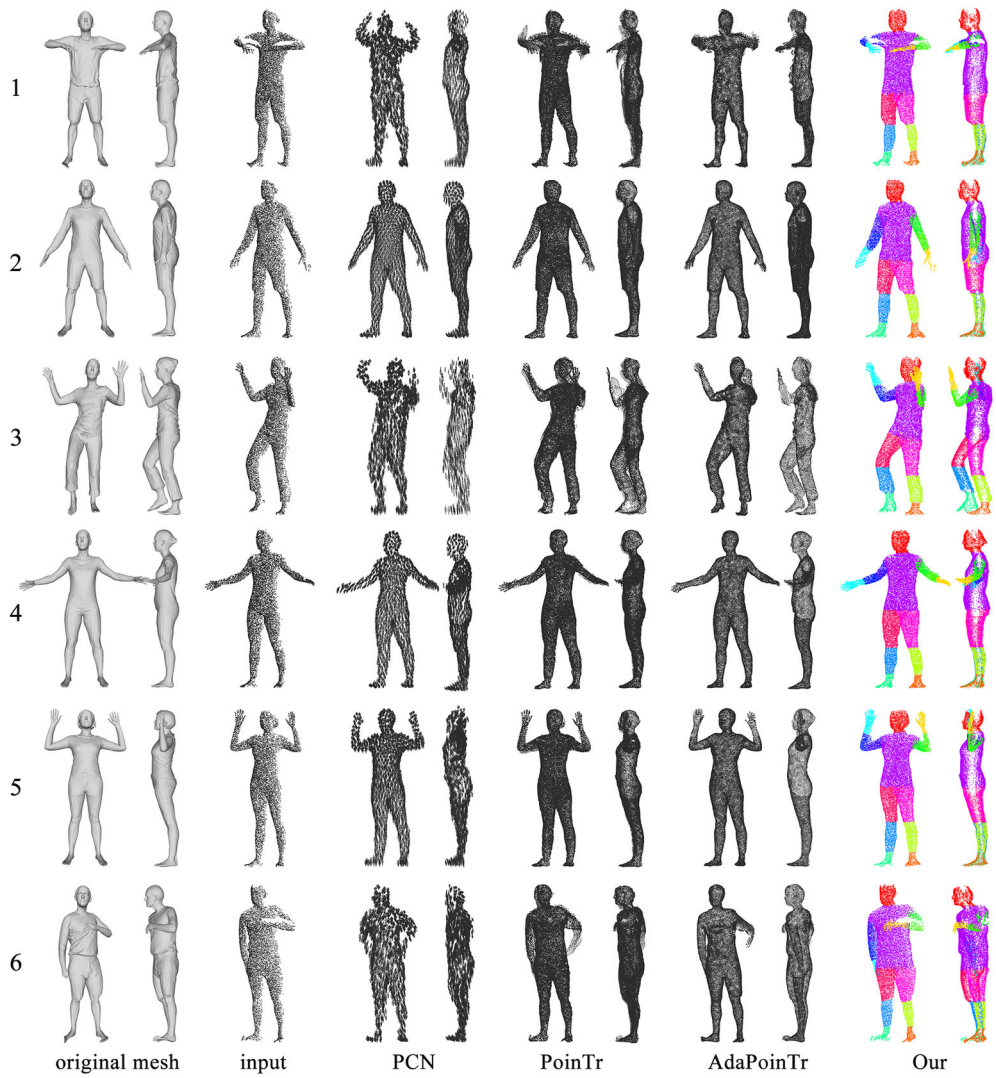

Fig. 9 Qualitative comparison of point cloud completion results among different methods

incomplete observations increase reconstruction ambiguity. Nonetheless, the part feature aggregation mechanism mitigates these issues by effectively distinguishing occluded from visible regions, enabling the model to maintain plausible global structure and achieve higher part-level accuracy even under partial input conditions.

6.6 Evaluation of the point cloud completion effect based on real scan data

To further verify the robustness and generalization capability of the proposed method in real-world scenarios, we conduct experiments on real human scans collected from multiple public datasets. The test set includes five representative samples: sample 1 and sample 2 from THuman2.0 [24], sample

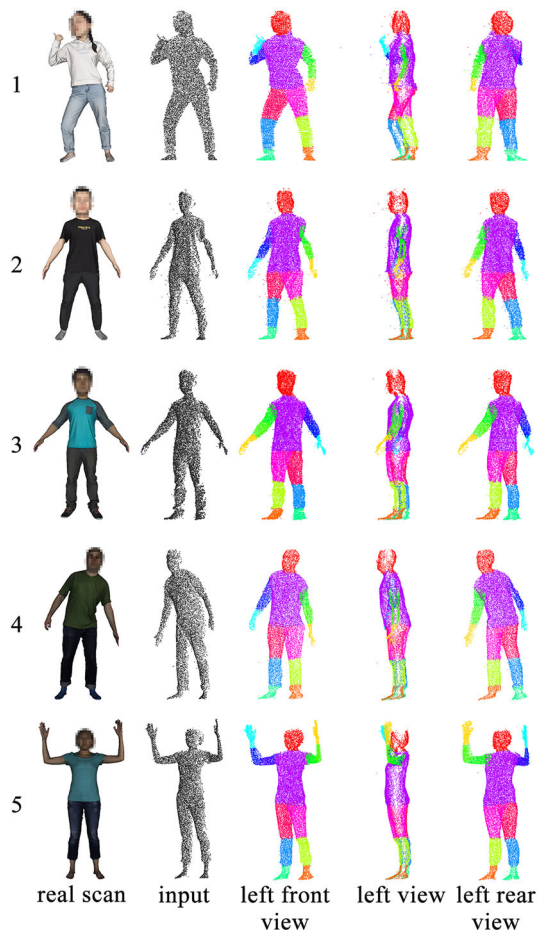

Fig. 10 Point cloud completion effect of real scan data

3 from Multi-Garment [26], and sample 4 and sample 5 from BUFF [25]. The completion results are presented in Fig. 10.

It should be noted that the input point clouds captured by depth cameras or RGB-D sensors exhibit missing regions due to self-occlusion, as well as noise, uneven sampling, and surface artifacts caused by sensor imperfections and clothing deformation. Despite these challenges, our method demonstrates strong robustness and generalization. It consistently reconstructs complete and structurally coherent human shapes under various poses, clothing types, and scanning conditions. The generated point clouds preserve geometric consistency and topological continuity, especially in complex regions such as limbs, armpits, and the back surface.

These results confirm the stability and practicality of the proposed method in real-world applications. Even with noisy and incomplete real scan inputs, the model maintains high-fidelity geometry and plausible global structure. This robustness makes it suitable for downstream tasks such as 3D virtual try-on, digital avatar creation, and immersive telepresence systems.

7 Conclusion

This paper presents a human point cloud completion method based on pose normalization, organized as a three-stage framework. The approach maps partial scans from arbitrary poses into the canonical space, infers missing back geometry, and restores the completed point cloud to the original space. By integrating the part feature aggregation mechanism and a single-branch offset prediction, the method improves both geometric accuracy and semantic consistency. Extensive experiments on synthetic and real datasets demonstrate that the proposed method outperforms state-of-the-art approaches in CD and EMD, showing strong robustness and generalization across complex poses and clothing conditions. The pose normalization effectively mitigates geometric inconsistency and enhances completion stability, while the single-branch offset inference ensures high computational efficiency.

Despite these advantages, several challenges remain before practical deployment. First, the quality of back reconstruction is still affected by severe self-occlusion and missing regions in the input, leading to uncertainty in unseen surfaces. Second, the unidirectional offset design limits the recovery of fine-scale geometric details such as fabric folds and local deformations. Third, performance may degrade when facing loose or layered clothing, where body and clothing interactions become highly non-rigid and difficult to model. Future work could combine parametric human fitting methods with spatial constraints between surface geometry and the underlying body to improve reconstruction accuracy. Expanding to larger and more diverse datasets would further enhance robustness and generalization.

Acknowledgements The authors did not receive support from any organization for the submitted work.

Author contributions L.M. conceptualized the research, developed and implemented the algorithm, and wrote the main manuscript text. L.G. provided research support and supervision. L.X. contributed to the research ideas and analysis. All authors reviewed the manuscript.

Data availability Data is provided within the manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Li, X., Li, G., Li, M., Cheng, X., Chen, F.: Design of undercloth-ing 3D human body measurement and reconstruction system for clothed individuals. Text. Res. J. (2025). https://doi.org/10.1177/00405175251345144
2. Tang, Y., Zhang, Q., Liu, Y., Hou, J.: Human as points: explicit point-based 3D human reconstruction from single-view rgb images.

IEEE Trans. Pattern Anal. Mach. Intell. (2025). https://doi.org/10.1109/TPAMI.2025.3552408
3. Lee, J., Cho, W., Jang, S., Lee, M., Shin, S.G., Han, S., Jo, H., Kim, S.: 3D pose estimation of workers in complex radioactive environments using a single RGB-D camera. Nucl. Eng. Technol. (2025). https://doi.org/10.1016/j.net.2025.103631
4. Zhou, M., Chen, J., Huang, X., An, P.: JointFormer: joint-enhanced 3D human point cloud completion based on transformer. APSIPA Trans. Signal Inf. Process. (2025). https://doi.org/10.1561/116.20240041
5. Wang, L., Li, J., Guo, S., Han, S.: A cascaded graph convolutional network for point cloud completion. Vis. Comput. **41**(1), 659–674 (2025). https://doi.org/10.1007/s00371-024-03354-x
6. Xiao, B., Da, F.: Three-stage generative network for single-view point cloud completion. Vis. Comput. **38**(12), 4373–4382 (2022). https://doi.org/10.1007/s00371-021-02301-4
7. Liu, Y., Sun, X., Shao, W., Yuan, Y.: S2ANet: combining local spectral and spatial point grouping for point cloud processing. Virtual Real. Intell. Hardw. **6**(4), 267–279 (2024). https://doi.org/10.1016/j.vrih.2023.06.005
8. Yuan, W., Khot, T., Held, D., Mertz, C., Hebert, M.: PCN: point completion network. In: 2018 International Conference on 3D vision (3DV), Verona, Italy, pp. 728–737 (2018). https://doi.org/10.1109/3DV.2018.00088
9. Wang, D., Tang, L., Zhu, L., Yang, Z.X.: Mutual information maximization based similarity operation for 3D point cloud completion network. IEEE Signal Process. Lett. **29**, 1217–1221 (2022). https://doi.org/10.1109/LSP.2022.3162139
10. Chang, Y., Jung, C., Xu, Y.: FinerPCN: high fidelity point cloud completion network using pointwise convolution. Neurocomputing **460**, 266–276 (2021). https://doi.org/10.1016/j.neucom.2021.06.080
11. Zhu, X., Yao, X., Zhang, J., Zhu, M., You, L., Yang, X., Zhang, J., Zhao, H., Zeng, D.: TMSDNet: transformer with multi-scale dense network for single and multi-view 3D reconstruction. Comput. Anim. Virtual Worlds (2024). https://doi.org/10.1002/cav.2201
12. Yu, X., Rao, Y., Wang, Z., Liu, Z., Lu, J., Zhou, J.: PoinTr: diverse point cloud completion with geometry-aware transformers. In: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), Electr network, pp. 12498–1250 (2021). https://doi.org/10.1109/ICCV48922.2021.01227
13. Yu, X., Rao, Y., Wang, Z., Lu, J., Zhou, J.: AdaPoinTr: diverse point cloud completion with adaptive geometry-aware transformers. IEEE Trans. Pattern Anal. Mach. Intell. **45**(12), 14114–14130 (2023). https://doi.org/10.1109/TPAMI.2023.3309253
14. Neverova, N., Güler, RA., Kokkinos, I.: Dense pose transfer. In: 2018 European Conference on Computer Vision (ECCV), Munich, Germany (2018). https://doi.org/10.1007/978-3-030-01219-9_8
15. Tan, F., Tang, D., Dou, m., Guo, K., Pandey, R., Keskin, C., Du, R., Sun, D., Bouaziz, S., Fanello, S., Tan, P., Zhang, Y.: HumanGPS: geodesic preserving feature for dense human correspondences. In: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Electr network, pp. 1820–1830 (2021). https://doi.org/10.1109/CVPR46437.2021.00186
16. Donati, N., Sharma, A., Ovsjanikov, M.: Deep geometric functional maps: robust feature learning for shape correspondence. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Electr network, pp. 8589–8598 (2020). https://doi.org/10.1109/CVPR42600.2020.00862
17. Ginzburg, D., Raviv, D.: Cyclic functional mapping: self-supervised correspondence between non-isometric deformable shapes. In: 2020 European Conference on Computer Vision (ECCV), Electr network, pp. 36–52 (2020). https://doi.org/10.1007/978-3-030-58558-7_3
18. Aygün, M., Lähner, Z., Cremers, D.: Unsupervised dense shape correspondence using heat kernels. In: 2020 International Conference on 3D Vision (3DV), Electr network, pp. 573–582 (2020). https://doi.org/10.1109/3DV50981.2020.00067
19. Roufosse, J., Sharma, A., Ovsjanikov, M.: Unsupervised deep learning for structured shape matching. In: 2019 IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, South Korea, pp. 1617–1627 (2019). https://doi.org/10.1109/ICCV.2019.00170
20. Lunscher, N., Zelek, J.: Deep learning whole body point cloud scans from a single depth map. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT (2018). https://doi.org/10.1109/CVPRW.2018.00157
21. Li, X., Li, G., Li, M., Song, H.: Parametric body reconstruction based on a single front scan point cloud. IEEE Trans. Vis. Comput. Graph. **31**(9), 5816–5828 (2025). https://doi.org/10.1109/TVCG.2024.3475414
22. Wang, J., Lu, X., Bennamoun, M., Sheng, B.: Non-rigid point cloud registration via anisotropic hybrid field harmonization. IEEE Trans. Pattern Anal. Mach. Intell. **47**(9), 7898–7915 (2025). https://doi.org/10.1109/TPAMI.2025.3572584
23. Ma, Q., Yang, J., Ranjan, A., Pujades, S., Pons-Moll, G., Tang, S., Black, M.: Learning to dress 3D people in generative clothing. In 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Electr network, pp. 6468–6477 (2020). https://doi.org/10.1109/CVPR42600.2020.00650
24. Yu, T., Zheng, Z., Guo, K., Liu, P., Dai, Q., Liu, Y.: Function4D: real-time human volumetric capture from very sparse consumer RGBD sensors. In: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Electr network, pp. 5742–5752 (2021). https://doi.org/10.1109/CVPR46437.2021.00569
25. Zhang, C., Pujades, S., Black, M., Pons-Moll, G.: Detailed, accurate, human shape estimation from clothed 3D scan sequences. In: 2017 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, pp. 5484–5493. https://doi.org/10.1109/CVPR.2017.582
26. Bhatnagar, B., Tiwari, G., Theobalt, C., Pons-Moll, G.: Multi-garment net: learning to dress 3D people from images. In: 2019 IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, South Korea, pp. 5419–5429. https://doi.org/10.1109/ICCV.2019.00552

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Ming Li is a Ph.D. student at the School of Mechatronic Engineering and Automation, Shanghai University, China. His research interests include computer aided design, computer vision, 3D human body reconstruction, and machine learning.

Guiqin Li is an Associate Professor at the School of Mechatronic Engineering and Automation, Shanghai University, China. She received her B.S., M.S., and Ph.D. degrees from Henan University of Science and Technology, Jilin University, and Northwestern Polytechnical University, respectively. Her research focuses on smart design and manufacturing, robotics, and multiphysics coupled simulation. She has published over 100 peer-reviewed papers in international journals and conferences and is a Senior Member of the Chinese Mechanical Engineering Society and a member of the Computer Graphics Special Committee of the China Graphics Society.

Xihang Li is a Lecturer at the College of Mechanical and Automotive Engineering, Ningbo University of Technology, China. His research interests include 3D human body reconstruction, computer vision, automated equipment, and intelligent manufacturing.